PROFINITE SPACES ARE EXACTLY THE HAUSDORFF QUASI-COMPACT TOTALLY DISCONNECTED SPACES, AND ARE COFILTERED LIMITS OF FINITE DISCRETE SPACES

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- The statement that an inverse limit of quasi-compact spaces is quasi-compact. Invoked inside the counted proof (source lines 4364-4365). Makes the limit of finite discrete spaces quasi-compact.
- The statement that in a quasi-compact Hausdorff space the connected component of a point is the intersection of its clopen neighbourhoods. Invoked inside the counted proof (source lines 4391-4393). Makes the comparison map injective.
- The statement that an intersection of closed subsets of a quasi-compact space is nonempty when all finite intersections are. Invoked inside the counted proof (source line 4394). Makes the comparison map surjective.
- The statement that a bijective map of quasi-compact Hausdorff spaces is a homeomorphism. Invoked at the end of the counted proof (source line 4395). Upgrades the bijection to a homeomorphism.

Source reference. Original lemma lemma-bijective-map: sources/stacks-topology/pinned-original-source.tex, source lines 3498–3498. Navigation pointer only: the counted text refers to the Stacks item lemma-bijective-map, which is an existing labeled result of this same chapter and is not part of this card. The pointer carries the target's own label as its number and its definition line as its span; its content is not reproduced here.

Source reference. Original lemma lemma-connected-component-intersection-compact-Hausdorff: sources/stacks-topology/pinned-original-source.tex, source lines 2078–2078. Navigation pointer only: the counted text refers to the Stacks item lemma-connected-component-intersection-compact-Hausdorff, which is an existing labeled

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Source reference. Original lemma lemma-intersection-closed-in-quasi-compact: sources/stacks-topology/pinned-original-source.tex, source lines 1941–1941. Navigation pointer only: the counted text refers to the Stacks item lemma-intersection-closed-in-quasi-compact, which is an existing labeled result of this same chapter and is not part of this card. The pointer carries the target's own label as its number and its definition line as its span; its content is not reproduced here.

Source reference. Original lemma lemma-inverse-limit-quasi-compact: sources/stacks-topology/pinned-original-source.tex, source lines 2692–2692. Navigation pointer only: the counted text refers to the Stacks item lemma-inverse-limit-quasi-compact, which is an existing labeled result of this same chapter and is not part of this card. The pointer carries the target's own label as its number and its definition line as its span; its content is not reproduced here.

Lemma 0.1. *Let X be a topological space. The following are equivalent*

(1) *X is a profinite space, and*
(2) *X is Hausdorff, quasi-compact, and totally disconnected.*

If this is true, then X is a cofiltered limit of finite discrete spaces.

Proof. Assume (1). Choose a diagram $i \mapsto X_i$ of finite discrete spaces such that $X = \lim X_i$. As each X_i is Hausdorff and quasi-compact we find that X is quasi-compact by Lemma lemma-inverse-limit-quasi-compact. If $x, x' \in X$ are distinct points, then x and x' map to distinct points in some X_i. Hence x and x' have disjoint open neighbourhoods, i.e., X is Hausdorff. In exactly the same way we see that X is totally disconnected.

Assume (2). Let $\mathcal{I}$ be the set of finite disjoint union decompositions $X = \coprod_{i \in I} U_i$ with U_i nonempty open (and closed) for all $i \in I$. For each $I \in \mathcal{I}$ there is a continuous map $X \to I$ sending a point of U_i to i. We define a partial ordering: $I \leq I'$ for $I, I' \in \mathcal{I}$ if and only if the covering corresponding to I' refines the covering corresponding to I. In this case we obtain a canonical map $I' \to I$. In other words we obtain an inverse system of finite discrete spaces over $\mathcal{I}$. The maps $X \to I$ fit together and we obtain a continuous map

$$X \longrightarrow \lim_{I \in \mathcal{I}} I$$

We claim this map is a homeomorphism, which finishes the proof. (The final assertion follows too as the partially ordered set $\mathcal{I}$ is directed: given two disjoint union decompositions of X we can find a third refining both.) Namely, the map is injective as X is totally disconnected and hence $\{x\}$ is the intersection of all open and closed subsets of X containing x (Lemma lemma-connected-component-intersection-compact-Hausdorff) and the map is surjective by Lemma lemma-intersection-closed-in-quasi-compact. By Lemma lemma-bijective-map the map is a homeomorphism. □